

PAGE 1: UNIFIED ENTERPRISE GATEWAY

SSO Multi-Tenant Login & Role-Based Access Architecture (Smart Login)

SPEC SHEET 01 / 06

ROLE SELECTION & MULTI-TENANT ISOLATION DESK

Master HQ Global Command (Patna HQ)

Access Level: SUPER_ADMIN (Omni-Branch Aggregation)

- Aggregates live data across Patna, Delhi, Mumbai & Kolkata
- Authorizes financial vouchers & indents exceeding Rs 50,000
- Exclusive capability to provision new regional branches
- Real-time P&L, workforce muster counts & active machinery

Scoped Site Branch Admin / Ground Ops

Access Level: BRANCH_ADMIN / STAFF (Scoped Node)

- Restricted strictly to WHERE branch_id = user.branch_id
- Real-time GPS geofenced muster roll (500m radius lock)
- Daily Progress Report (DPR) submission & defect snags
- Local petty cash & stock indents queue below Rs 50k

Security & Trust Safeguards (Pre-Established Backend Architecture):

- Multi-Tenant Guard: Database queries enforce multi-branch tenant isolation per request.
- JWT Session Continuity: High-security signed tokens contain user role and branch authorization.
- Anti-Spoof GPS Telemetry: GPS coordinates verified against Haversine radius to prevent muster spoofing.
- Microcopy Standard: GSTIN 10BIEPD2766D2ZX verified; ISO 9001:2015 compliant data validation.

AUTHENTICATION & GATEWAY LOGIC

```
// POST /api/auth/login
const user = await authenticate(email, password);

if (user.role === "SUPER_ADMIN") {
  // Global Multi-Site View
  return router.push("/dashboard/super-admin");
} else if (user.role === "BRANCH_ADMIN") {
  // Scoped Site View
  return router.push("/dashboard/branch");
} else {
  // Staff Tasks View
  return router.push("/dashboard/employee");
}
```

```
// Global Status Bar Feed:
Workforce: 842/910 Present (92.5%)
Payroll: Rs 4,82,500 Disbursed Today
Pending HQ Approvals: 6 Desks
```

PAGE 2: FRONT-PAGE DESIGN SPECIFICATION

Exact Layout as image_37.png with Dynamic 5-Image Background Slider & Role CTAs

SPEC SHEET 02 / 06

JMK Engineering & Developers | Services Projects Quality Tenders

[+ Add Branch] (HQ) | [Contact Supplier / Request RFQ]

EST. 2021 | PROPRIETORSHIP FIRM | ISO COMPLIANT OPERATIONS

Engineering the Future of Infrastructure & Turnkey Construction

[Explore Product Catalog] [Request Instant Quote / RFQ]

5 DISTINCT GEOMETRIC SHAPED PRODUCT SPEC CARDS (image_37.png exact):

[Keystone Shape]

Expansion Joints

SS Joint 8.20mm

PDF [Download]

[Teardrop Shape]

Shuttering Plate

20kg / 27mm

PDF [Download]

[Circle Shape]

Centering Sheets

MS 27kg / 13kg

PDF [Download]

[Organic Shape]

Centerg Plates

POT PTFE 10k kN

PDF [Download]

[Hexagon Shape]

H-Frame Scaffold

Props Box 30mm

PDF [Download]

SLIDER & ROLE CTA CALLOUTS

1. DYNAMIC 5-IMAGE LOOPING SLIDER:

- Image 1: Heavy Infrastructure Casting (Concrete pier)
- Image 2: Industrial Site Management Hub (Warehouse)
- Image 3: GPS + Face ID Biometric Attendance Kiosk
- Image 4: Material Logistics Hub (Steel racks)
- Image 5: Multi-Branch Command Room Data Visualization

2. TRANSITION LOGIC:

- Automatic cross-fade duration: 1.0s
- Slide display hold time: 5.0s
- Smooth GSAP & CSS transform rendering

3. ROLE-BASED CONDITIONAL CTA:

- if (user.role === "SUPER_ADMIN") -> [+ Add Branch]
- if (user.role === "BRANCH_ADMIN") -> [Approvals Desk]
- Standard Contact Supplier / Request RFQ action

PAGE 3: SUPER ADMIN HQ COMMAND HUB

Pan-India Real-Time Multi-Site Telemetry, Budget Allocation & Approvals Matrix

SPEC SHEET 03 / 06

Total Operational Sites 4 Regional Hubs Patna HQ, Delhi, Mumbai, Kolkata	Workforce Present Today 842 / 910 Staff 92.5% GPS Verified In-Fence	Monthly Budget Allocated Rs 1.75 Crore Spend: Rs 94.8 Lakhs (54%)	Pending HQ Approvals 6 Desks Active > Rs 50,000 Authority Tier
Patna Central Works (PAT-HQ) Workforce: 312 Workers Production: 48 MT Today DPR Status: Live CNC Punch Active GPS Geofence: Active Verified Action: [Deep-Dive Audit]	Delhi NCR Depot (DEL-NCR) Workforce: 198 Workers Production: 32 MT Staged DPR Status: Strip Seal Joint Lot GPS Geofence: Active Verified Action: [Deep-Dive Audit]	Mumbai Logistics Node (BOM-WST) Workforce: 174 Workers Production: 24 MT Deployed DPR Status: POT-PTFE Proof Test GPS Geofence: Active Verified Action: [Deep-Dive Audit]	Kolkata Metro Depot (CCU-EST) Workforce: 158 Workers Production: 18 MT Erected DPR Status: H-Frame Falsework Setup GPS Geofence: Active Verified Action: [Deep-Dive Audit]

PAGE 4: ONBOARDING & PROVISIONING MODAL

Workflow for Deploying New Regional Branches & Geofenced Logistics Nodes

SPEC SHEET 04 / 06

PROVISION NEW REGIONAL LOGISTICS HUB / SITE

Endpoint: POST /api/branches | Role Authority: Super Admin Patna HQ Only

Branch Name

Ahmedabad Western Corridor Depot

Branch Code (3-4 Chars)

AMD-WST

Central GPS Coordinates

Lat: 23.0225° N, Lng: 72.5714° E

Geofence Verification Radius

500 Meters (High Precision Anti-Spoof)

Allocated Monthly Budget

Rs 35,00,000 / Month

Assigned Regional Manager

Er. Amit Trivedi (Senior Structural Engineer)

ACTION: [PROVISION & DEPLOY NEW HUB]

PAGE 5: SCOPED BRANCH WORKSPACE & GPS GEOFENCE

Scoped Site Command Desk & Anti-Spoof Live Attendance Radius Architecture

SPEC SHEET 05 / 06

DELHI NCR LOGISTICS DEPOT (DEL-NCR) - SCOPED VIEW

Incharge: Er. Vikram Sharma | Site Address: IMT Manesar Sector 8, Gurgaon

GPS GEOFENCE LIVE RADIAL RADAR (500M PERIMETER)

Site Center: 28.3642° N, 76.9421° E

Live Worker Status: 198 Workers Inside Fence (94.2%) | 12 Out of Fence Exceptions

Auto-Action: Out-of-fence punch requires Branch Admin OTP / Photo Override

Local Shift Output: 32 Metric Tonnes Staged | Inventory: 1,420 Centering Sheets

Daily Progress Report (DPR): Submitted & Verified by Site QA/QC

SCOPED ARCHITECTURE SECURITY

1. TENANT ENFORCEMENT:

Every query in the Branch Admin session is bound:
WHERE branch_id = "br_delhi_hub"

2. ZERO CROSSOVER:

Branch managers cannot see competitor depot rates
or raw material costs from Patna HQ.

3. HARDWARE GPS TELEMETRY:

Uses Mobile HTML5 Geolocation + Cell Tower ID
Maximum acceptable accuracy: +/- 15m radius.

PAGE 6: DIGITAL MUSTER ROLL & RA BILLING

Automated Daily Labour Waging, Voucher Queues & Subcon Settlements

SPEC SHEET 06 / 06

DAILY DIGITAL MUSTER ROLL (842 Active Today)

Ramesh Paswan (Senior Fitter)

Hub: Patna Central Works | GPS: Verified Inside (114m)

Wage: Rs 850 / Day

Sunil Yadav (Arc Welder)

Hub: Delhi NCR Depot | GPS: Verified Inside (42m)

Wage: Rs 950 / Day

Anil Shinde (Rigging Lead)

Hub: Mumbai Logistics Hub | GPS: Verified Inside (88m)

Wage: Rs 900 / Day

Bikram Roy (QA Inspector)

Hub: Kolkata Metro Node | GPS: Verified Inside (210m)

Wage: Rs 1,100 / Day

Mukesh Kumar (CNC Operator)

Hub: Patna Central Works | GPS: Verified Inside (55m)

Wage: Rs 950 / Day

HQ FINANCIAL APPROVAL DESK (Vouchers > Rs 50k)

SAIL Structural Steel Channel Lot

Origin: Delhi Hub • Category: Material / Maintenance

Rs 1,45,000

250 MT Hydraulic Rig Overhaul

Origin: Patna Works • Category: Material / Maintenance

Rs 68,000

POT-PTFE Bearing Rubber Compound Batch

Origin: Mumbai Hub • Category: Material / Maintenance

Rs 2,10,000

Hot-Dip Zinc Ingot Raw Supply

Origin: Patna Works • Category: Material / Maintenance

Rs 3,25,000